Manual Chapter : Configuring an SSL Intercept Explicit Proxy Mode

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Load Balancing Passive Mode FTP Traffic with Data Channel Optimization

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Implementing APM System Authentication

Configuring an SSL Intercept Explicit Proxy Mode

Configuring an Explicit HTTP Proxy Chain

Manipulating HTTPS Traffic by Using a Third-Party Device

Legal Notices

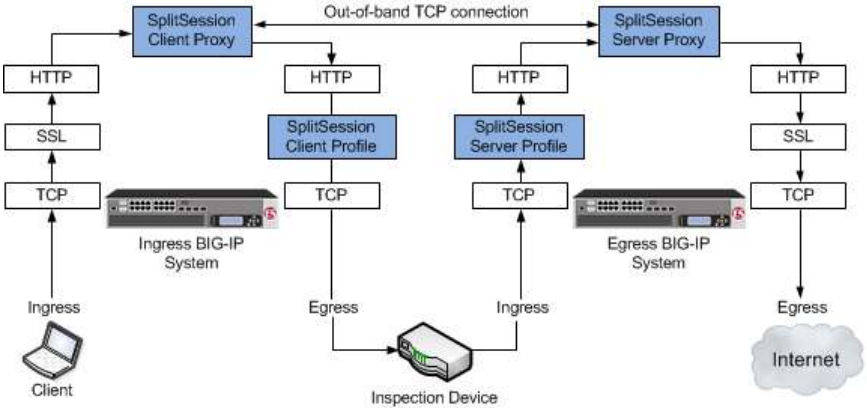
Configuring an SSL Intercept Explicit Proxy Mode

About SSL intercept explicit proxy mode

A typical SSL intercept explicity proxy mode configuration includes two BIG-IP devices, one configured to manage half-proxy client traffic and one configured to manage half-proxy server traffic. When the ingress BIG-IP system receives a client request, SSL decrypts the request. The ingress BIG-IP system then sends metadata to the egress BIG-IP system by means of the out-of-band TCP connection and sends the request data to the inspection device. When the egress BIG-IP system receives the metadata through the out-of-band connection and the request from the inspection device, it uses the information in the metadata, re-encrypts the request, and forwards it to the destination server.

The following illustration depicts an example configuration.

An example SSL intercept explicity proxy mode configuration



The SplitSession Client profile type

The SplitSession Client profile defines the client parameters in an SSL intercept explicit proxy mode configuration. This profile enables you to configure a Peer Port, which specifies the port for the SplitSession peer that is connected to the out-of-band connection, and the Peer IP address, which specifies the IP address for the SplitSession peer that is connected to the out-of-band connection.

The SplitSession Server profile type

The SplitSession Server profile defines the server parameters in an SSL intercept explicit proxy mode configuration. This profile enables you to configure a Listen Port, which specifies the port that the SplitSession server listens on for the out-of-band connection, and the Listen IP address, which specifies the IP address that the SplitSession server listens on for the out-of-band connection.

Task summary for configuring SSL intercept explicit proxy mode

Complete these tasks to configure an SSL intercept explicit proxy configuration.

Creating a pool to process HTTP traffic for an inspection device

You can create a pool that includes an inspection device to process HTTP requests.

1. On the Main tab, click **Local Traffic › Pools .**
   The Pool List screen opens.
2. Click **Create.**
   The New Pool screen opens.
3. In the **Name** field, type a unique name for the pool.
4. Using the **New Members** setting, add each resource that you want to include in the pool:
   a. Type an IP address in the **Address** field.
   b. Type `80` in the **Service Port** field, or select **HTTP** from the list.
   c. (Optional) Type a priority number in the **Priority** field.
   d. Click **Add.**
5. Click **Finished.**

The new pool appears in the Pools list.

Creating an ingress explicit proxy virtual server

Before you configure an ingress explicit proxy virtual server, you need to configure a SplitSession Client profile and pool to assign to the virtual server.

You can configure an ingress explicit proxy virtual server to manage the client split-session half-proxy traffic from a client to the inspection device.

1. On the Main tab, click **Local Traffic › Virtual Servers .**
   The Virtual Server List screen opens.
2. Click **Create.**
   The New Virtual Server screen opens.
3. In the **Name** field, type a unique name for the virtual server.
4. In the **Description** field, type a description of the virtual server.
5. In the **Source Address** field, type `0.0.0.0/0` for the source address and prefix length.
6. In the **Destination Address/Mask** field, type an IP address in CIDR format.

   The supported format is address/prefix, where the prefix length is in bits. For example, to select all IP addresses, an IPv4 address/prefix is `0.0.0.0/0`, and an IPv6 address/prefix is `::/0`. To specify a network, an IPv4 address/prefix is `10.07.0.0` or `10.07.0.0/24`, and an IPv6 address/prefix is `ffe1::/64` or `2001:ed8:77b5::/64`. When you use an IPv4 address without specifying a prefix, the BIG-IP system automatically uses a `/32` prefix.

   For best results, F5 recommends that you enter the subnet that matches your destination server network.

7. In the **Service Port** field, type `443` or select **HTTPS** from the list.
8. From the **HTTP Profile** list, select **http.**
9. For the **SSL Profile (Client)** setting, select a client SSL profile.
10. From the **Protocol** list, select **TCP.**
11. From the SplitSession Client Profile list, select **splitsessionclient** or a custom SplitSession Client profile.
12. From the **Default Pool** list, select the name of the HTTP server pool that you previously created.
13. Click **Finished.**

An ingress explicit proxy virtual server is configured to manage the client split-session half-proxy traffic from a client to the inspection device.

Creating a SplitSession Server profile

You can create a SplitSession Server profile to define the server parameters in an SSL intercept explicit proxy mode configuration.

1. On the Main tab, click **Local Traffic › Profiles › Other › SplitSession Server** .
   The SplitSession Server profile list screen opens.
2. Click **Create**.
   The New SplitSession Server Profile screen opens.
3. In the **Name** field, type a unique name for the profile.
4. From the **Parent Profile** list, retain the default value or select another existing profile of the same type.
5. In the **Listen Port** field, type a value for the the port of the SplitSession server listens on for the out-of-band connection.
6. In the **Listen IP** field, type the IP address of the SplitSession server listens on for the out-of-band connection.
7. Click **Finished**.

A SplitSession Server profile to define the server parameters in an SSL intercept explicit proxy mode configuration is available to assign to a virtual server.

Creating a custom Server SSL profile

Create a custom server SSL profile to support SSL forward proxy.

1. On the Main tab, click **Local Traffic › Profiles › SSL › Server** .
   The Server SSL profile list screen opens.
2. Click **Create**.
   The New Server SSL Profile screen opens.
3. In the **Name** field, type a unique name for the profile.
4. For **Parent Profile**, retain the default selection, **serverssl**.
5. From the **Configuration** list, select **Advanced**.
6. Select the **Custom** check box.
   The settings become available for change.
7. From the **SSL Forward Proxy** list, select **Enabled**.

   You can update this setting later, but only while the profile is not assigned to a virtual server.
8. From the **SSL Forward Proxy Bypass** list, select **Enabled** (or retain the default value **Disabled**).

   The values of the **SSL Forward Proxy Bypass** settings in the server SSL and the client SSL profiles specified in a virtual server must match. You can update this setting later but only while the profile is not assigned to a virtual server.
9. Scroll down to the **Secure Renegotiation** list and select **Request**.
10. Click **Finished**.

The custom Server SSL profile is now listed in the SSL Server profile list.

Creating a pool to manage HTTPS traffic

You can create a pool (a logical set of devices, such as web servers, that you group together to receive and process HTTPS traffic) to efficiently distribute the load on your server resources.

1. On the Main tab, click **Local Traffic › Pools** .
   The Pool List screen opens.
2. Click **Create**.
   The New Pool screen opens.

3. In the **Name** field, type a unique name for the pool.
4. For the **Health Monitors** setting, assign **https** or **https_443** by moving it from the **Available** list to the **Active** list.
5. From the **Load Balancing Method** list, select how the system distributes traffic to members of this pool.

   The default is **Round Robin**.
6. For the **Priority Group Activation** setting, specify how to handle priority groups:
   - Select **Disabled** to disable priority groups. This is the default option.
   - Select **Less than**, and in the **Available Members** field type the minimum number of members that must remain available in each priority group in order for traffic to remain confined to that group.
7. Use the **New Members** setting to add each resource that you want to include in the pool:
   a. In the **Address** field, type an IP address.
   b. In the **Service Port** field type `443` , or select **HTTPS** from the list.
   c. (Optional) Type a priority number in the **Priority** field.
   d. Click **Add**.
8. Click **Finished**.

The HTTPS load balancing pool appears in the Pool List screen.

Creating an egress explicit proxy virtual server

Before you configure an egress explicit proxy virtual server, you need to configure a SplitSession Server profile and pool to assign to the virtual server.

You can configure an egress explicit proxy virtual server to manage the server split-session half-proxy traffic from an inspection device to a server.

1. On the Main tab, click **Local Traffic** › **Virtual Servers** .
   The Virtual Server List screen opens.
2. Click **Create**.
   The New Virtual Server screen opens.
3. In the **Name** field, type a unique name for the virtual server.
4. In the **Description** field, type a description of the virtual server.
5. In the **Source Address** field, type `0.0.0.0/0` for the source address and prefix length.
6. In the **Destination Address/Mask** field, type an IP address in CIDR format.

   The supported format is address/prefix, where the prefix length is in bits. For example, to select all IP addresses, an IPv4 address/prefix is `0.0.0.0/0`, and an IPv6 address/prefix is `::/0`. To specify a network, an IPv4 address/prefix is `10.07.0.0` or `10.07.0.0/24`, and an IPv6 address/prefix is `ffe1::/64` or `2001:ed8:77b5::/64`. When you use an IPv4 address without specifying a prefix, the BIG-IP system automatically uses a `/32` prefix.

   For best results, F5 recommends that you enter the subnet that matches your destination server network.
7. In the **Service Port** field, type `443` or select **HTTPS** from the list.
8. For the **SSL Profile (Server)** setting, select a server SSL profile.
9. From the **Protocol** list, select **TCP**.
10. From the SplitSession Server Profile list, select **splitsessionserver** or a custom SplitSession Server profile.
11. From the **Default Pool** list, select the name of the HTTP server pool that you previously created.
12. Click **Finished**.

An egress explicit proxy virtual server is configured to manage the server split-session half-proxy traffic from an inspection device to a server.

Creating a SplitSession Client profile

You can create a SplitSession Client profile to define the client parameters in an SSL intercept explicit proxy mode configuration.

1. On the Main tab, click **Local Traffic › Profiles › Other › SplitSession Client .**
   The SplitSession Client profile list screen opens.
2. Click **Create.**
   The New SplitSession Client Profile screen opens.
3. In the **Name** field, type a unique name for the profile.
4. From the **Parent Profile** list, retain the default value or select another existing profile of the same type.
5. In the **Peer Port** field, type a value for the the port of the SplitSession peer assigned to the out-of-band connection.
6. In the **Peer IP** field, type the IP address of the SplitSession peer assigned to the out-of-band connection.
7. Click **Finished.**

A SplitSession Client profile to define the client parameters in an SSL intercept explicit proxy mode configuration is available to assign to a virtual server.

Creating a custom Client SSL profile

You create a custom Client SSL profile when you want the BIG-IP system to terminate client-side SSL traffic for the purpose of:

- Authenticating and decrypting ingress client-side SSL traffic
- Re-encrypting egress client-side traffic

By terminating client-side SSL traffic, the BIG-IP system offloads these authentication and decryption/encryption functions from the destination server.

1. On the Main tab, click **Local Traffic › Profiles › SSL › Client .**
   The Client SSL profile list screen opens.
2. Click **Create.**
   The New Client SSL Profile screen opens.
3. In the **Name** field, type a unique name for the profile.
4. Select **clientssl** in the **Parent Profile** list.
5. From the **Configuration** list, select **Advanced.**
6. Select the **Custom** check box.
   The settings become available for change.
7. Next to Client Authentication, select the **Custom** check box.
   The settings become available.
8. From the **Configuration** list, select **Advanced.**
9. Modify the settings, as required.
10. Click **Finished.**

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